

Sentiment Analysis Using NLP and Machine Learning

Final Project Report

1. Introduction

The objective of this project is to develop a sentiment analysis system capable of classifying text data into Negative, Neutral, and Positive categories. Sentiment analysis plays a crucial role in understanding public opinion, customer feedback, and social media trends. This project applies Natural Language Processing (NLP) techniques and machine learning algorithms to extract meaningful insights from textual data.

2. Dataset Description

The dataset consists of preprocessed text data stored in the column 'clean_text' and corresponding sentiment labels stored in the column 'category'. The sentiment labels are encoded as -1 (Negative), 0 (Neutral), and 1 (Positive). Rows containing missing values were removed to ensure data quality.

Class distribution analysis revealed a moderate class imbalance, with Positive sentiment being the majority class, followed by Neutral and Negative sentiments.

3. Feature Extraction

Text data cannot be directly used by machine learning models. Therefore, feature extraction was performed using the Term Frequency–Inverse Document Frequency (TF-IDF) technique. TF-IDF converts text into numerical vectors by assigning higher weights to words that are frequent in a document but rare across the corpus. Both unigrams and bigrams were used to capture contextual information.

4. Model Building

Two supervised machine learning models were implemented for sentiment classification: Multinomial Naive Bayes and Support Vector Machine (SVM). Naive Bayes was used as a baseline model, while SVM with class-weight balancing was used as the primary model due to its effectiveness in high-dimensional text classification tasks.

5. Model Evaluation – Naive Bayes

The Naive Bayes model achieved an overall accuracy of approximately 75.8%. While the model showed high recall for the Positive class, it struggled to correctly identify Negative sentiments, resulting in lower recall for the minority class. This behavior is typical of probabilistic models when class distributions are imbalanced.

6. Model Evaluation – Support Vector Machine (SVM)

The Support Vector Machine model with balanced class weights demonstrated superior performance. The confusion matrix showed strong diagonal values, indicating accurate classification across all three sentiment classes. The model effectively reduced bias toward the majority class and improved the identification of Negative and Neutral sentiments.

The confusion matrix results indicate that most samples were correctly classified, with relatively few misclassifications occurring between adjacent sentiment classes such as Neutral and Positive.

7. Visualization and Insights

Data visualization techniques such as sentiment distribution plots, confusion matrices, and word clouds were used to interpret model performance and sentiment trends. Word cloud analysis helped identify

frequently occurring words associated with each sentiment category, providing qualitative insights into the dataset.

8. Conclusion

This project successfully demonstrates the application of NLP and machine learning techniques for sentiment analysis. TF-IDF proved to be an effective feature extraction method, and the SVM classifier delivered reliable performance across all sentiment classes. The system can be extended further using deep learning models such as LSTM or transformer-based architectures like BERT for enhanced contextual understanding.